



THE NEXT EVOLUTION OF OUR GEOSPATIAL 3D TWIN

Receipts, not Renders.

Making our climate digital twin **believable** – grounded in how the world's most-trusted city twins actually earn belief.

*"Our unfair advantage isn't fidelity – we're barred from it.
It's honesty made visible."*

DATE

9 August 2026

PREPARED FOR

The co-founders

FOCUS

Credibility & the raise

SOURCES

Primary-verified

01 • EXECUTIVE SUMMARY

The twin is good. Now we make it **believable**.

We have built a working urban-heat digital twin of three Kolkata wards from nothing but free, open, legally-clean data – under constraints that would stop most teams (no drone capture over Indian cities, no photorealistic tiles, no field sensors, no first-world budget). It looks great. That is exactly the problem.

The belief gap. "Looks great" is what a demo looks like. Climate scientists trust **rigor**; investors trust **defensibility**. We already have the rigor – leave-one-overpass-out validation, bootstrap confidence intervals, 50 Landsat overpasses, a daytime correction we **rejected** because it was worth 0.003 K, a height check we published as "underpowered" rather than overclaim. But almost none of it is visible in the product. A visitor sees a beautiful surface and one caveat chip. The moat is real and invisible.

The plan. Make the rigor **legible** – turn every number in the twin into something you can interrogate. Click a reading, get its "receipt": source, acquisition date, method, and validated accuracy. This is precisely what the world's most-trusted city and geospatial platforms do, and what most impressive-looking twins skip.

Why it wins. It is cheap (it surfaces work already done – no new science, no chasing precision), it is uniquely ours (honest measured-vs-modelled disclosure is a differentiator, not a liability), and it converts our biggest constraint into our brand: **we can't fake it, so we prove it**.

Why it funds us. It is also the fastest route to a raise. Climate-adaptation funders and grant panels back **defensibility, not demos** – the "is this real?" test is what kills most applicants, and our receipts moat answers it on sight. That positions us for a founder-friendly, blended path: non-dilutive grants + a climate accelerator to prove it, then a climate-VC seed (\$07-08).

THE ASK

Endorse a two-phase build of a "provenance / receipts" layer on the twin – Phase 1 (shippable now) surfaces per-layer and per-reading receipts; Phase 2 adds the measured-vs-modelled distinction and an in-product validation dossier. No changes to the model itself.

02 • WHERE WE STAND — AND THE CONSTRAINT THAT DEFINES US

The constraint isn't a weakness. It's the strategy.

What we've built. A browser-based 3D urban-heat instrument on a live map: ~3,500 real building footprints per ward, extruded with measured heights, a live physics heat simulation forced by live weather, an intervention model priced in real ₹, and a resilience score — running at 60 fps, degrading gracefully down to a budget phone. Every layer comes from open data: Microsoft & Google Open Buildings (footprints + heights), ECOSTRESS and Landsat (land-surface temperature), Sentinel-2 (vegetation), OpenStreetMap (roads), on an open MapLibre basemap.

What we cannot do. We cannot fly drones over Indian cities, and photogrammetric/photoreal 3D tiles don't cover our scattered wards uniformly (and carry runtime-only licensing + per-session billing we won't put on a public page). So we build **procedurally**, from satellites and open footprints — a modelled twin, not a captured one.

Peer-reviewed critique of the field warns that many city twins are "dashboards that look impressive but lack depth."^[9] **Fidelity is not credibility.** Being barred from fidelity forced us toward the thing that actually earns trust.

The honest read. Our model's accuracy is already close enough for the decisions the tool supports — the structural gap is not precision, it is **provenance**. The right move is not to chase a tenth of a degree; it is to make what we already know defensible, visible, and interrogable. This briefing is how.

03 · THE CORE INSIGHT

Honesty, shown correctly, buys integrity.

There is a counter-intuitive trap in showing uncertainty, and getting it right is the whole game.

A 2026 controlled study (N = 161) found that visualising uncertainty on maps generally reduces viewers' trust – but the reduction hits their confidence in the **data's accuracy**, while barely touching their perception of the **mapmaker's integrity**. Critically, low-uncertainty displays carried **no** trust penalty versus showing nothing at all.^[1]

Shout your error bars and people trust the data less. Make uncertainty quiet, optional, and always paired with the source that explains it – and you are read as rigorous, not unreliable.

DESIGN CONSEQUENCE ADOPTED THROUGHOUT THIS PLAN

This single finding shapes the entire design:

- **Lead with the confident surface.** The best-estimate heat map is the default view.
- **Uncertainty is one click away, never in your face.** Confidence intervals and quality flags live inside the receipt, next to the provenance that justifies them.
- **Candour is the feature.** Stating a limit plainly – as the EU's own flagship heat service does^[2] – reads as science. Hiding it reads as marketing.

Our competitors in "impressive demos" won't do this, because it takes real work and real data. That is precisely why it is defensible.

04 • OTHER DIGITAL TWINS – AND THE PHILOSOPHY THAT EARNS TRUST

Read for their philosophy, not their graphics.

The world's most-trusted city and geospatial twins are each built on an explicit philosophy about **why** they exist and **how** they earn belief – and it is almost never "make it look real." We read them for that philosophy, and for what we can adopt. Every claim below is primary-verified (see Sources).

Destination Earth – Urban Heat Island service • EU (ECMWF • ESA • EUMETSAT)

HEAT • BENCHMARK

The EU's flagship Earth twin runs VITO's physics-based UrbClim model at 100 m. Its credibility comes from validation-in-the-open: UrbClim's pan-European track record was validated against **67 meteorological stations**, and the DestinE demonstration separately reports that its results held up "certainly given the fact that most of these stations were in non-optimal measurement locations."^[2]

STEAL → Surface our validation the same way: report the error, name the limits out loud. Candour about non-ideal conditions is what reads as science.

Google Earth Engine – Data Catalog • the receipts gold standard

RECEIPTS

Every dataset carries a structured card: provider, temporal coverage, spatial resolution, bands/variables, linked license, a suggested citation, and a DOI.^[3] It is the reason scientists trust it as a working instrument.

STEAL → Copy this field set verbatim for every one of our layers (Open Buildings, ECOSTRESS, Landsat, Sentinel-2, OSM).

Our World in Data • measured vs. derived, made explicit

HONESTY

Every series is labelled by its measured source versus "Our World in Data based on..."; and any transformation is flagged; the tooling is open-sourced for scrutiny.^[4]

STEAL → Draw our biggest honest line – which pixels are satellite-measured vs. model-interpolated. This is our largest exposure and our strongest signal.

NYC Heat Vulnerability Index • operational definitions

PUBLIC-HEALTH AUTHORITY

States its exact trigger ("2 or more days with a heat index of 95 °F, or 1 or more days reaching 100 °F"), a clear 1-5 scale, and links the peer-reviewed method.^[5]

STEAL → Publish our exact heat metrics + method as an "About this instrument" page – no hand-waving.

What the flagships teach about trust and scale.

Virtual Singapore · NRF + Dassault Systèmes (3DEXPERIENCECity)

NATIONAL · SEMANTIC

Singapore's national twin is **semantic and analytical**, not merely visual — "a realistic and integrated 3D model with semantics and attributes,"^[6] built on the Land Authority's aerial-photo + airborne-LiDAR national mapping. A ~S\$73 million, five-year national R&D programme (NRF-attributed).

STEAL → Semantics = credibility. Our buildings already carry attributes; expose them. (Their proprietary platform + authoritative LiDAR is the part we *can't* match — we cite the posture, not the stack.)

Helsinki 3D+ / Kalasatama · City of Helsinki

OPEN DATA

Best-in-class **open** municipal twin: the city publishes both a semantic CityGML model and a photogrammetric reality mesh as open data under **CC BY 4.0**^[7] (the Kalasatama pilot mesh was built from over 42,000 aerial photos at ~7.5 cm). Proof that open data can be authoritative.

STEAL → Our whole stack is open by necessity — make that a virtue: publish per-layer data + the derivation method. Reproducibility is free credibility.

Digital Twin Victoria · Victorian Government + CSIRO, Australia

PROVENANCE-AS-PRODUCT

Backed by a record **AUD \$37.4M / 4-year** investment, its public value is a shared catalogue of **4,000+ datasets** — each carrying metadata, licensing (**CC BY 4.0**) and attribution, served through OGC-standard web services. Provenance treated as the product, at state scale.^[8]

STEAL → Our provenance manifest is a small version of exactly this — one structured source of truth for "where did this come from," per layer, per city.

UK Gemini Principles · CDBB, 2018

The canonical values framework for trustworthy twins — nine principles in three groups: **Purpose** (public good, value creation, insight), **Trust** (security, **openness**, quality), **Function** (federation, curation, evolution).^[10]

STEAL → Adopt "openness" + "quality" as stated values on our About page.

Cesium / OGC 3D Tiles · open standards

3D Tiles — introduced by Cesium in 2015, an OGC Community Standard since 2019 — plus Khronos's glTF (now ISO/IEC 12113) let one pipeline serve many cities.^[11]

STEAL → Keep the procedural build standards-clean so a ward pipeline drops onto any city.

THE SHARED PHILOSOPHY

Across every one of them the through-line is identical: **a twin earns trust by being open, traceable, and honest about its limits** — provenance treated as a public good, not polish mistaken for proof. That philosophy is ours to adopt in full, and it is exactly the one most "impressive" twins never commit to.

05 • WHAT WE NEED TO DO

A "receipts" layer: every number, interrogable.

The spine. A single provenance manifest – one structured source of truth for every layer and for the modelled surface: provider, dates, resolution, license, citation, method, and validated accuracy. The validation figures are [read from our accuracy engine, not copy-pasted](#), so a receipt can never disagree with our published numbers. New cities get their own manifest – it is portable by design.

Progressive disclosure – one system, three depths, so the same artifact serves a live demo, a self-serve skeptic, and an investor data-room:

GLANCE
Authority at rest

Persistent "data as of" readout, live-forcing time, satellite epoch, and a measured/modelled key – traceable without a click.

SUMMARY
The receipt

Click a layer or reading → source · acquisition date · method · validated accuracy, in the Earth-Engine card shape.

DOSSIER
Survive scrutiny

Drill in → the validation campaign, the confidence intervals, the honest limits. The full defensibility trail, in-product.

Phase 1 – shippable now

- Per-layer provenance cards (Earth-Engine field set) + the persistent "data as of" furniture.
- Per-reading receipts on the building card and the ward mean, with the phase-specific confidence band shown [quiet and on-demand](#).
- No model changes – this surfaces work already done.

Phase 2 – the depth that makes a scientist nod

- The measured-vs-modelled distinction drawn across the surface (inputs measured; surface modelled + validated).
- Source-native quality flags (e.g. ECOSTRESS's own best/good/suspect/poor grade^[12]) – the satellite's confidence, not one we invent.
- A quiet "how we know" dossier: leave-one-overpass-out, the confidence intervals, and the corrections we honestly ruled out.

06 · WHY THIS BEATS THE ALTERNATIVES – AND WHAT WE DELIBERATELY WON'T DO

The cheapest, highest-belief move on the board.

THE TEMPTING ALTERNATIVE	WHY WE'RE NOT DOING IT
Chase photoreal fidelity	Barred by regulation/budget – and fidelity ≠ credibility; the literature is explicit that impressive-looking twins are routinely discounted for lacking depth. ^[9]
Chase more model precision	Accuracy is already close enough for the decisions supported; another tenth of a degree changes no one's mind. The gap is legibility, not precision.
Ship louder dashboards	More gauges ≠ more trust. Interrogability does.

Anti-patterns we will avoid (each observed in the field)

- **Loud, always-on uncertainty** – reduces trust in the data itself.^[1] Ours is quiet and on-demand.
- **Provenance buried in a PDF** – even rigorous heat-mappers hide methodology in separate linked reports.^[13] Ours is one click from the number.
- **Over-claiming "digital twin"** – the term implies live, bidirectional physical coupling; scientists discount static visualisations that claim it.^[9] In-product we are precise: a **modelled urban-heat observatory, calibrated to satellite**. ("Digital twin" stays for the deck, not the methods page.)

THE THROUGH-LINE

Every credible benchmark we studied wins on **traceability and candour**, not graphics. We are constitutionally suited to that game – we were never allowed to play the other one.

07 · HOW WE SERIOUSLY SECURE FUNDING

The credibility moat is the fundraising strategy.

We are in the funded lane at the funded moment: ~71% of climate-adaptation capital flows to software/data, and risk-monitoring platforms were the fastest-growing adaptation category in H1 2025.^[15] And the #1 diligence risk is "is this real?" – 74% of climate-AI claims are unproven.^[16] Our honest, out-of-sample-validated twin answers that on sight. So we run a founder-friendly, blended path: non-dilutive money + an accelerator to prove it, then a climate-VC seed.

The money map – ranked for a pre-seed Indian climate-software team

TRACK	BEST-FIT AVENUES (PRIMARY-VERIFIED)	WHAT IT GIVES
Non-dilutive · India	DPIIT → SISFS (₹20L grant + ₹50L conv.) ^[17] · Climate Pitch (equity-free ₹20L, explicit adaptation/urban-resilience theme) ^[18] · Social Alpha Techtonic (₹35L urban pilot) ^[19] · NIDHI-SSS (₹1cr)	Grant / soft capital, little-to-no equity
Cloud + data credits	MS Founders Hub (\$150k, no equity) · NVIDIA Inception (equity-free) · AWS Activate (\$100k) · Google for Startups (\$200k) ^[20]	12–18 mo runway; avoids the Earth Engine commercial-licence trap
Accelerators	Third Derivative (equity-free, India-heavy) ^[21] · Katapult Climate (~8%, adaptation in-scope) · T-Hub AIC (India, pilot access)	Network + a credibility stamp
Climate / adaptation VC	Avaana · Transition VC · Ankur · Speciale (India) ^[22] · Lightsmith (adaptation, growth-stage – later)	Pre-seed / seed equity, on traction + moat
Dev-finance & philanthropy	ADB Challenges (≤ \$450k, startups eligible) ^[23] · Grand Challenges India (\$200k; funded a Karnataka heatwave project) ^[24]	Non-dilutive; theme-gated, watch windows

THE ONE MOVE THAT UNLOCKS THE MOST

Get **DPIIT recognition** (free, online) → join an incubator that is simultaneously a **SISFS incubator + a DST-NIDHI TBI + an Atal Incubation Centre**. One admission opens SISFS, NIDHI and AIC support at once – the grant engine for year one. Known dead-ends (so we don't waste cycles): BIRAC = biotech-only; MeitY GENESIS excludes Tier-I cities (Kolkata); GCF / Adaptation Fund only via an accredited entity (NABARD); USAID defunct; ESA / NASA / EU grants are India-ineligible without an EU entity.

08 · THE PLAN – PROOF, THEN SEED

A 12–18 month path from proof to seed.

- **Months 0–3 · Set the base (all non-dilutive).** DPIIT recognition; ship Phase-1 receipts + publish the [validation dossier](#); claim NVIDIA Inception + cloud credits (\$150k+ runway) on Copernicus / Planetary Computer; apply to a multi-accredited incubator.
- **Months 3–9 · Prove it.** SISFS + NIDHI + Climate Pitch; a global accelerator for the stamp (Third Derivative / Katapult); land **1–2 municipal or academic pilots** as traction; Phase-2 receipts (measured-vs-modelled + validation) become the defensibility artifact.
- **Months 9–18 · Raise the seed.** Walk into climate-VC rooms (Avaana / Transition / Ankur / Speciale; ADB Ventures) with a validated product, real pilots and grant wins already on the board – the moat doing the de-risking.

What funders actually want – and why we clear the bar

- **The right municipal owner + a specific, priced pilot/LOI** beats any waitlist – heat risk is owned by disaster/health/planning officers who never say "adaptation".^[25]
- **Validation is the moat, not the data** – free EO is commodity; our calibration, ground-truth and published accuracy are not.^[16] This is our whole thesis, and it is exactly what diligence now probes hardest.
- **Impact in beneficiary terms** – people protected, heat-exposure reduced – reported per deployment, alongside revenue. Segment funders: catalytic/grant first, commercial VC second.

B2G WHITESPACE TO EXPLOIT

There is **no Chief Heat Officer in any Indian city** – helping stand up the first is a ready-made champion.^[26] Warmest doors: the **Chennai Resilience Center** (CRO at IIT-Madras) and **NIUA's Climate Centre for Cities**. And a sharp, sourced pitch: Kolkata is a C40 member but **absent from C40's Cool Cities Network** – "you're C40, but not in its heat network; here's your Heat Action Plan, operationalised."

A commercial wedge worth naming: parametric heat-insurance is already live in our launch geography (SEWA · Swiss Re · ICICI Lombard, Gujarat), but it triggers on crude single-station / ERA5 data and its partners **openly admit the basis risk** – our validated, high-resolution, night-and-humidity-aware surface is the fix, making us a data-input vendor to underwriters, not only to cities.^[26]

Grants fund the proof · the proof wins the seed · the seed funds the scale – and every city we add widens the story without widening the cost.

DECISION REQUESTED

Green-light Phase 1 (receipts) and the DPIIT + incubator move in parallel. Together they cost no model code, surface rigor we already have, and turn the twin into something a scientist can interrogate and a funder can bank.

09 · SOURCES — DIGITAL TWINS & METHOD

Every claim, traceable.

Consistent with the thesis, this document carries its own receipts. Every source below was ✓ **PRIMARY-VERIFIED** — confirmed against its primary source on 9 Aug 2026, by three independent research passes.

#	SOURCE	STATUS
1	Trust & uncertainty-visualisation study (N=161) — arxiv.org/abs/2602.00248	✓ VERIFIED
2	Destination Earth — Urban Heat Island / UrbClim — destination-earth.eu	✓ VERIFIED · two validations distinguished
3	Google Earth Engine — Data Catalog dataset description — developers.google.com	✓ VERIFIED
4	Our World in Data — FAQ / methodology & labelling — ourworldindata.org/faqs	✓ VERIFIED
5	NYC Heat Vulnerability Index — a816-dohbesp.nyc.gov	✓ VERIFIED
6	Virtual Singapore — Dassault Systèmes press release + SLA (2014) + GovTech	✓ VERIFIED · S\$73M NRF-attributed
7	Helsinki 3D city model (CityGML + reality mesh, CC BY 4.0) — hel.fi · hri.fi	✓ VERIFIED
8	Digital Twin Victoria (Vic. Gov + CSIRO; \$374M/4yr; 4,000+ datasets; CC BY 4.0) — d.vic.gov.au · data.vic.gov.au	✓ VERIFIED
9	Digital-twin critique — Frontiers in Big Data, 10.3389/fdata.2023.1236397 — frontiersin.org	✓ VERIFIED
10	UK Gemini Principles — CDBB (Dec 2018) — cdbb.cam.ac.uk	✓ VERIFIED · verbatim
11	3D Tiles (Cesium 2015 → OGC Community Standard 2019) + glTF (Khronos, ISO/IEC 12113) — cesium.com · ogc.org · khronos.org	✓ VERIFIED
12	ECOSTRESS L2T LSTE v002 quality flags — lpdaac.usgs.gov	✓ VERIFIED
13	CAPA Strategies — Heat Watch data/methodology — capastrategies.com	✓ VERIFIED
14	Data provenance/quality standards — Data Nutrition Project (datanutrition.org); ISO 19115 lineage / ISO 19157 quality	✓ VERIFIED

Delta internal evidence referenced (leave-one-overpass-out validation, bootstrap CIs, 50-overpass Landsat campaign, ICESat-2 height check) is documented in the project's own accuracy engine and validation records. **Funding sources overleaf.**

10 • SOURCES — FUNDING

The funding map, sourced.

Compiled 9 Aug 2026 across nine independent research passes against primary/official pages. Programme windows and cheque sizes rotate – reconfirm the live application window on each portal before applying.

#	SOURCE
15	Adaptation is the funded lane (~71% of adaptation capital to software/data; resilience fastest-growing H1-2025) – tailwindfutures.com · ctvc.co
16	The "is-this-real" bar – 74% of climate-AI claims unproven; moat = validation not data – greenwebfoundation.org · latitudemedia.com
17	Startup India Seed Fund (SISFS) + DPIIT recognition (the gateway); DST NIDHI-SSS (₹1cr) / PRAYAS (₹10L) – seedfund.startupindia.gov.in · startupindia.gov.in · nidhi.dst.gov.in
18	Climate Collective – Climate Pitch (equity-free ₹20L, adaptation/urban-resilience) – climatepitch.org · climatecollective.net
19	Social Alpha – Tectonic: Sustainable Urban Transition (₹35L pilot) – socialalpha.org
20	Cloud credits – NVIDIA Inception · MS Founders Hub · AWS Activate · Google for Startups. Open EO backbones (free, commercial): Cope rnicus · MS Planetary Computer
21	Accelerators – Third Derivative (equity-free) third-derivative.org · Katapult Climate katapult.vc
22	India climate VC (seed leg) – Avaana · Transition VC · Ankur · Speciale Invest – avaana.capital . Academic-grant route: Wellcome Climate Impacts (£2.5M, via an India academic host) – wellcome.org
23	ADB Challenges (grants ≤ ~\$450k, startups eligible) – challenges.adb.org
24	Grand Challenges India (BIRAC + Gates; funded a Karnataka heatwave-resilience project) – gcgh.grandchallenges.org
25	What funders actually want – LOIs / paid pilots beat waitlists; the adaptation impact-metric framework – tran.vc · msci-institute.com
26	Parametric heat-insurance wedge (basis-risk) – SEWA · Swiss Re · ICICI Lombard, Gujarat – inclusivecapitalism.com · downtoearth.org.in · demand-side: World Bank India cities (Kolkata, UHI, 2025)
27	B2G doors – Arsht-Rock Heat Action Platform onebillionresilient.org · Resilient Cities Network (Chennai) resilientcitiesnetwork.org · NIUA Climate Centre for Cities niua.in/c-cube · C40 Cool Cities c40.org

Do-not-chase (verified dead-ends): USAID (defunct 2025) · BIRAC (biotech-only) · MeitY GENESIS (excludes Tier-I cities) · ESA / NASA / EU direct grants (India-ineligible without an EU entity) · GCF & Adaptation Fund (only via an accredited entity such as NABARD) · carbon credits for cooling (no methodology exists).